

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

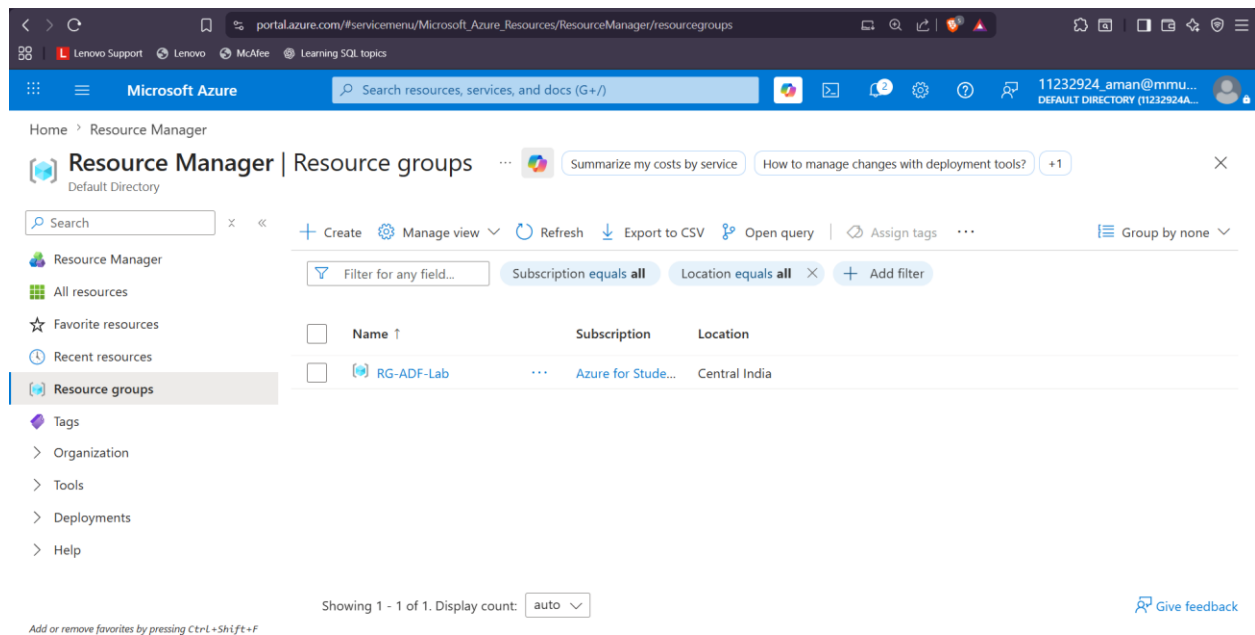
1.**Objective:** Understand Azure cloud concepts and build an end-to-end data pipeline using Storage Account and Azure Data Factory

2.Steps and the Screenshots of Exceution

Step 1: Resource Group Creation

Explanation:

A Resource Group named **RG-ADF-Lab** was created to organize and manage all Azure resources used in this project.



Step 2: Storage Account Creation

Explanation:

An Azure Storage Account was created to store the dataset and other files required for the data pipeline.

Home

adflabstorage123_1781423148900 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

- Inputs
- Outputs
- Template

Deployment is in progress

Deployment name : adflabstorage123_1781423... Start time : 6/14/2026, 1:17:23 PM
 Subscription : Azure for Students Correlation ID : 9e1193c5-435d-475e-b244...
 Resource group : RG-ADF-Lab

Deployment details

Resource	Type	Status
adflabstorage123	Storage account	Conflict

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Step 3: Blob Containers Creation

Explanation:

Two Blob Containers were created:

- **source-data** for storing the original dataset.
- **destination-data** for storing the copied dataset.

Home > Storage center | Blob Storage > adflabstorage123

adflabstorage123 | Containers

Storage account

data

+ Add container Upload Refresh Delete Change access level Restore containers Edit columns

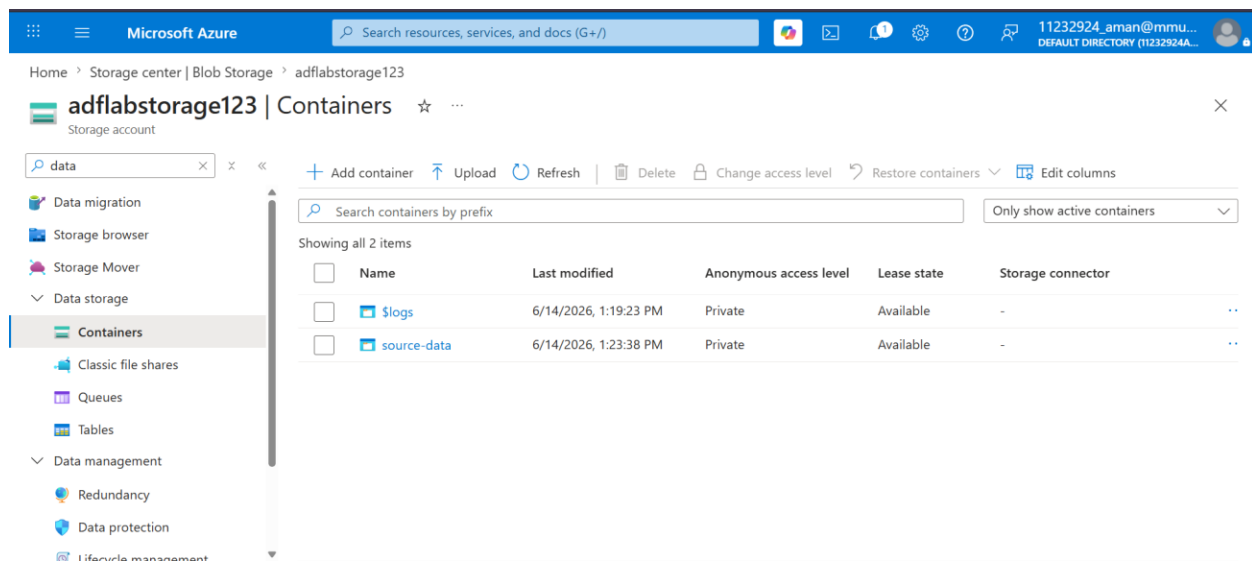
Search containers by prefix

Only show active containers

Showing all 3 items

	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	\$logs	6/14/2026, 1:19:23 PM	Private	Available	-
☐	destination-data	6/14/2026, 1:25:43 PM	Private	Available	-
☐	source-data	6/14/2026, 1:23:38 PM	Private	Available	-

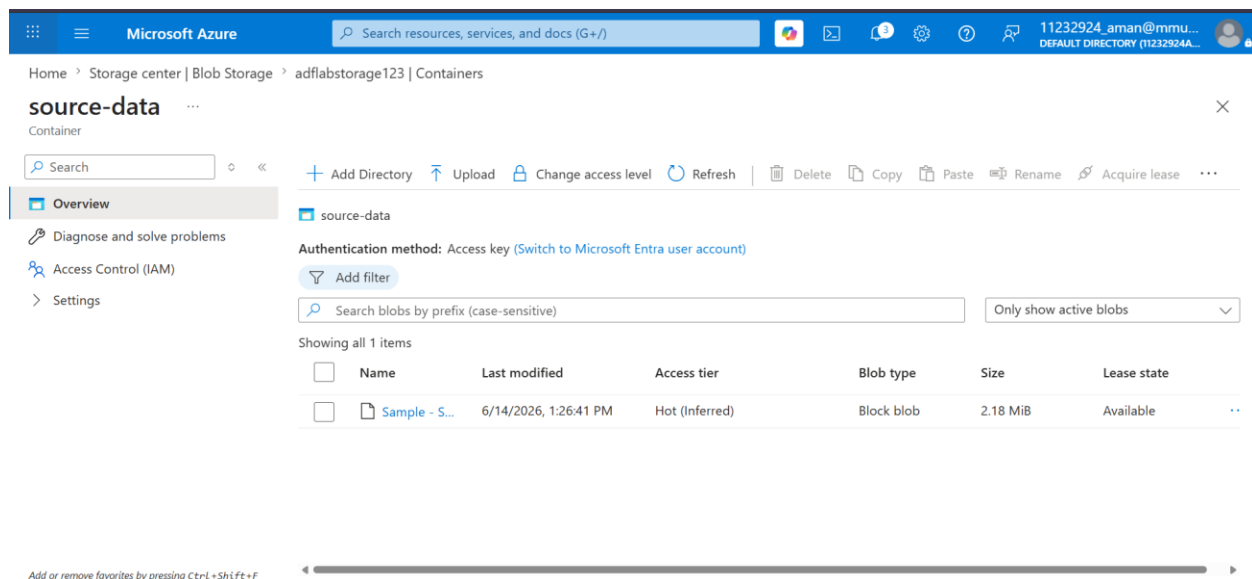
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Step 4: Dataset Upload

Explanation:

The Superstore CSV dataset was uploaded to the **source-data** container for processing through Azure Data Factory.



Step 5: Azure Data Factory Creation

Explanation:

Azure Data Factory (ADF) was created to build, manage, and monitor the data pipeline.

Microsoft Azure

Search resources, services, and docs (G+/)

11232924_aman@mmu...
DEFAULT DIRECTORY (11232924A...)

Home

CreateDataFactory-20260614132734 | Overview

Deployment

Search

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Overview

- Inputs
- Outputs
- Template

✓ Your deployment is complete

Deployment name : CreateDataFactory-20260... Start time : 6/14/2026, 1:30:00 PM
Subscription : [Azure for Students](#) Correlation ID : 1c7a449f-6cba-4333-a468...
Resource group : [RG-ADF-Lab](#)

Deployment details

Resource	Type	Status
NIKHIL-Lab	Data factory (V2)	OK

Next steps

[Go to resource](#)

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Step 6: ADF Studio Home

Microsoft Azure

Data Factory > NIKHIL-Lab

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11232924_aman@mmu...
DEFAULT DIRECTORY

Home

Data factory

NIKHIL-Lab

New

- Ingest**
Copy data at scale once or on a schedule.
- Orchestrate**
Code-free data pipelines.
- Transform data**
Transform your data using data flows.
- Configure SSIS**
Manage & run your SSIS packages in the cloud.

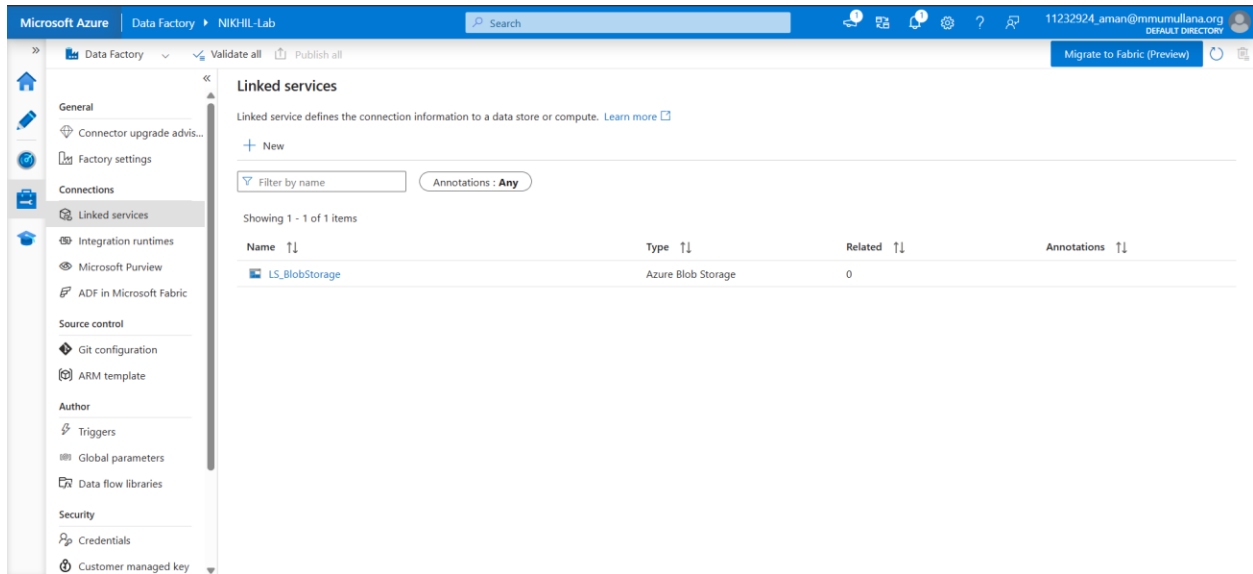
Recent resources

No items to show
Your recently opened resources will show up here.

Step 7: Linked Service Configuration

Explanation:

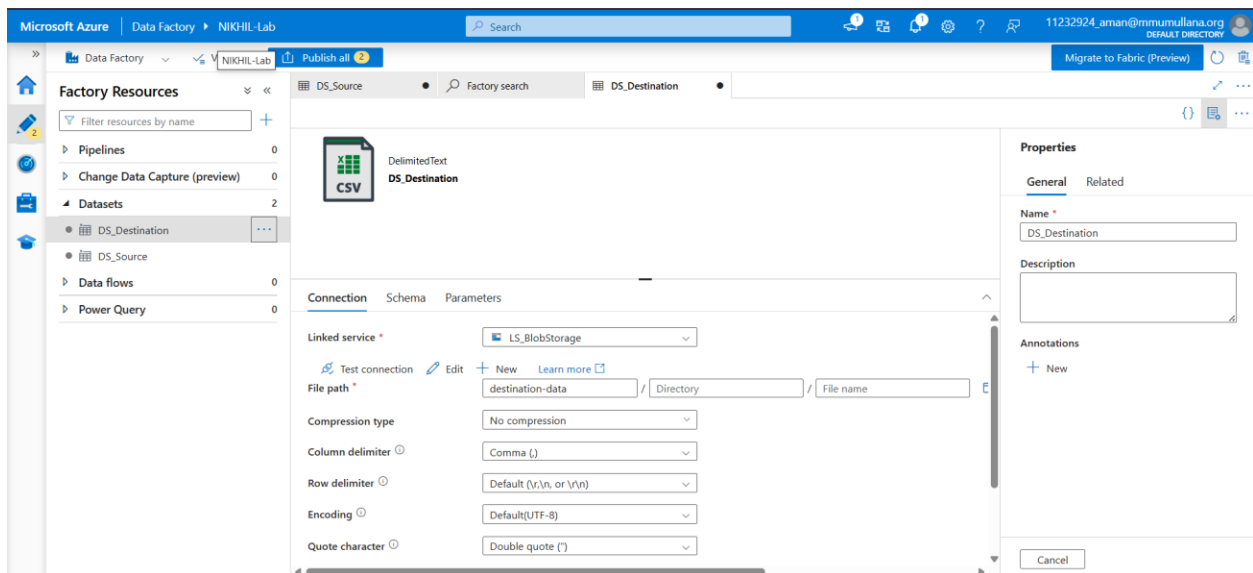
A Linked Service was configured to establish a connection between Azure Data Factory and Azure Blob Storage.



Step 8: Dataset Creation

Explanation:

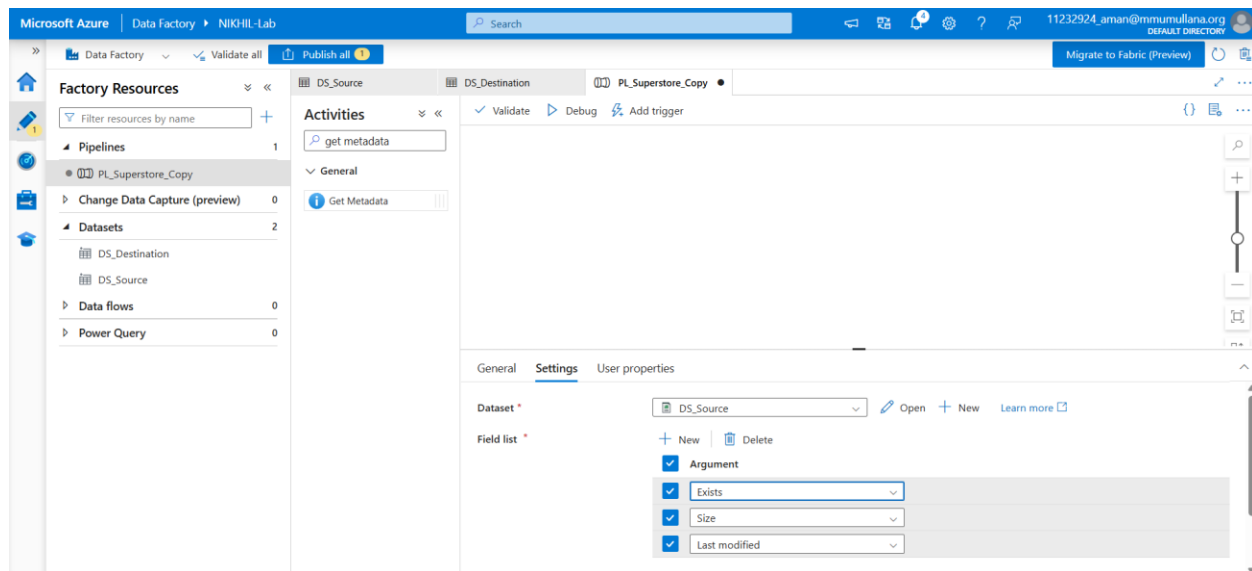
Source and destination datasets were created in ADF to define the input and output locati



Step 9: Get Metadata Activity

Explanation:

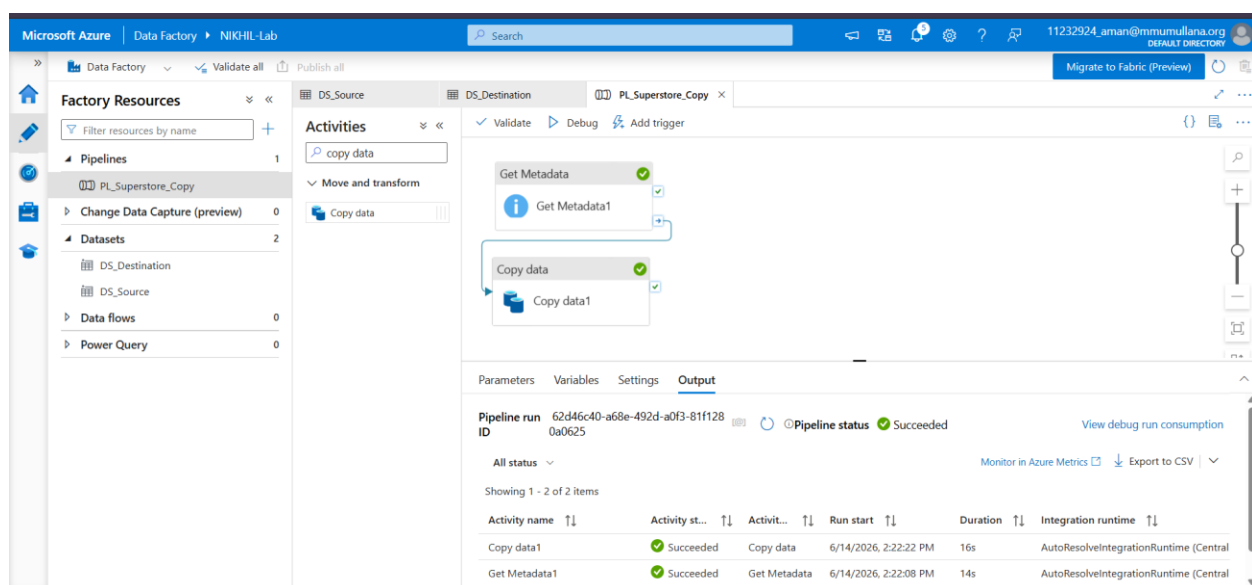
The Get Metadata activity was used to retrieve file details such as file name, file size, and last modified date.



Step 10: Pipeline Development

Explanation:

A pipeline was developed using Get Metadata and Copy Data activities to validate and transfer the dataset.



Step 11: IAM Role Assignment

Explanation:

Reader and Contributor roles were assigned to provide secure access between Azure Data Factory and Azure Storage Account.

These explanations are short, simple, and suitable for a Word report or internship submission.

Search

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data console

Download role assignments

Add

Download role assignments

Edit columns

Refresh

Delete

Feedback

Name	Type	Role	Scope	Condition
Owner (2)				
☐ Aman kumar 732b3d70-8173-4...	User	Owner	Subscription (Inherit...	None
☐ Aman kumar 732b3d70-8173-4...	User	Owner	Subscription (Inherit...	None
Reader (1)				
☐ Aman kumar 732b3d70-8173-4...	User	Reader	This resource	None
Container Registry Contributor and Data Access Configuration Administrator (1)				
☐ Aman kumar 732b3d70-8173-4...	User	Container Registry Contrib...	This resource	None

Showing 1 - 4 of 4 results.